NutriAI Technical Brief

By Raine Man Wing Yu

NutriAI is an AI-powered recommendation system that aims to generate a personalized 7-day 3 meal plan for a user based on their clinical conditions, allergens, dietary preference, and nutrient targets. It was generated based on an offline USDA FoodData Central food snapshot of 10,400 deduced food records. To evaluate its effectiveness, NutriAI has been tested on 4 personas with different conditions, needs, and preferences.

This brief document elaborates on the three BAX 423 techniques that were chosen and integrated in the pipeline, the architecture, the clinical persona results, and the limitations.

1. Chosen Techniques and Why

a. Embeddings and FAISS nearest-neighbor retrieval (Lecture 5)

Each food item in NutriAI contains several nutrients including calories, fiber, sugar, sodium, iron, calcium, vitamin B12, vitamin D, and zinc. Because there are many nutrients to consider, embeddings are an ideal technique as it attributes each food item to a numerical vector that summarizes each food's nutritional value. Rather than being compared by the name of the food item, it will be compared mathematically. FAISS is then used after embeddings to search through the 10,400 food items and retrieve the foods that are most similar to a user's goals.

b. Deep Neural Network Ranking (Lecture 7)

The deep neural network ranking is used to assign a personalized score to each food item based on their nutrients. Foods that have sugar, sodium, or saturated fats will receive lower scores whereas foods that have vitamins, fiber, and protein will receive higher scores. This technique learns from the relationship between nutrients and user profile and their dietary restrictions to ensure that users receive recommendations that cater to their profile. The benchmark testing, specifically the goal_fit column, shows that it produced better matches than the FAISS-only approach, making it an ideal system.

technique	goal_fit	diversity	latency_ms
dnn	0.734	1.049	786.5
faiss	0.675	0.892	4.025

c. Thompson-sampling bandit – reinforcement learning (Lectures 8-9)

Thompson-sampling bandit is used to enhance personalization from learning from user feedback. Even if users have the same health conditions, they may still have different food preferences. Thus, the feedback can help the unfavored meals appear less frequently and the favored meals to appear more frequently to a user. A benefit to using this technique is that it does not need to be retrained often and it does not need large amounts of data to make personalized recommendations. It is an efficient system.

2. System Architecture

NutriAI uses a step-by-step process to generate meal plans. Food items that do not align with a user's preferences and clinical conditions are immediately excluded and are explained as to why they are removed. The remaining foods are then ranked and evaluated into a personalized plan. The final recommendation ensures that the food items are nutritionally appropriate to the user.

Stage	**Function**	**What it does**
1. Intake & validation	Persona dataclass	Collect age, gender, dietary preferences, allergies, health conditions, and calorie goals.
2. Dedup	NutriEngine.__init__	Remove duplicate food names and standardize food records so each food is represented consistently.
3. Clinical filter	clinical_filter	IBS → remove high-FODMAP + garlic/onion/wheat;
		GERD → exclude citrus/tomato/fried/spicy/caffeine/chocolate + fat > 20g;
		T2 → GI ≤ 55;
		Hypertension → sodium > 120mg/100g screen.
4. Allergy exclusion	allergy_filter	Remove foods containing allergens or intolerances such as dairy, gluten, soy, or tree nuts.
5. Dietary filter	diet_filter	Vegan / vegetarian / pescatarian / non-veg tag match, plus optional no-pork.
6. Food Ranking	rank_faiss / rank_weighted	Rank foods that best match the user's nutritional and dietary goals
7. Food Diversity	diversify (MMR)	Same foods are not repeatedly recommended and meal diversity is maximized throughout the week.

8. Meal Plan Generation	build_plan + pick_csp/greedy	Generate a 7-day meal plan that satisfies calorie targets, dietary restrictions, and nutrient requirements. Portion sizes align with calorie and sodium goals
9. RDA analysis	analyze / rda_for	Compare the meal plan based on recommended nutrient guidelines and flag any deficiencies
10. Feedback	apply_feedback	Save meal/replace meal to update the bandit arm and the ranker weights

3. Benchmark

The benchmark evaluates the top 20 recommended foods for each persona based on three factors (goal-fit, diversity, and latency). These results are reported from benchmark_results.csv.

Persona	**Technique**	**Goal-fit**	**Diversity**	**Latency**
Priya	FAISS	0.636	1.035	4.1
Priya	DNN	0.707	1.05	308.1
Ravi	FAISS	0.653	0.916	3.5
Ravi	DNN	0.689	1.045	1314.5
Mei	FAISS	0.725	0.795	4.2
Mei	DNN	0.746	1.051	756.6
James	FAISS	0.686	0.823	4.3
James	DNN	0.794	1.051	766.8

4. Persona Results – Safety Pass/Fail

Each meal plan was tested using 4 personas who had different conditions and dietary preferences. The result shows that the meal plans generated aligns with all requirements, avoids allergens, meets nutritional targets, and more. The results reported are taken directly from persona_results.csv.

Persona	Profile	Avg kcal	Diversity	Fish	Result

Priya–IBS, vegetarian, lactose-intolerant	Zero high-FODMAP (no garlic/onion/wheat); zero dairy/lactose; vegetarian-only; iron ≥80% RDA/day; diversity ≥0.70	1840	0.707	0	PASS
Ravi– GERD, non-veg, gluten-free	No GERD triggers (citrus/tomato/fried/spicy/caffeine/chocolate, fat>20g); no gluten/wheat (strict celiac oat rule); no pork; B12 ≥80% RDA/day	2203	0.755	11	PASS
Mei–T2 diabetes, vegan, tree-nut allergy	Every food GI ≤55; vegan-only; no tree-nut allergen; fiber ≥25g/day; diversity ≥0.70	1610	0.747	0	PASS
James–hypertension, pescatarian, soy allergy	Daily sodium ≤1500mg; ≥3 fish/seafood meals/week; no soy allergen; pescatarian-only; potassium ≥80% RDA/day	2029	0.782	3	PASS

Result: 4/4 personas pass without triggering any violation. These results demonstrate the system's ability to consistently produce safe, personalized, and nutritionally appropriate meal plans across a variety of dietary and clinical scenarios. All four also achieve above 0.70 diversity.

5. System & Project Limitations

a. Simplified Food Database: The offline snapshot of USDA FoodData Central does not contain real-world branded food products. Some food categories are assigned using rule-based methods, so they could be categorized falsely
b. Simplified Health Rules: Conditions and guidelines are simplified and may not reflect professional medical advice
c. Limited Nutrient Scope: NutriAI does not cover all nutrients found in a food item
d. Challenges for Highly Constrained Diets: users with highly restricted diets may have fewer options to choose from NutriAI's recommendation
e. Fixed Meal Plan: NutriAI generates only breakfast, lunch, and dinner and does not account for snacks or meal changes that could occur